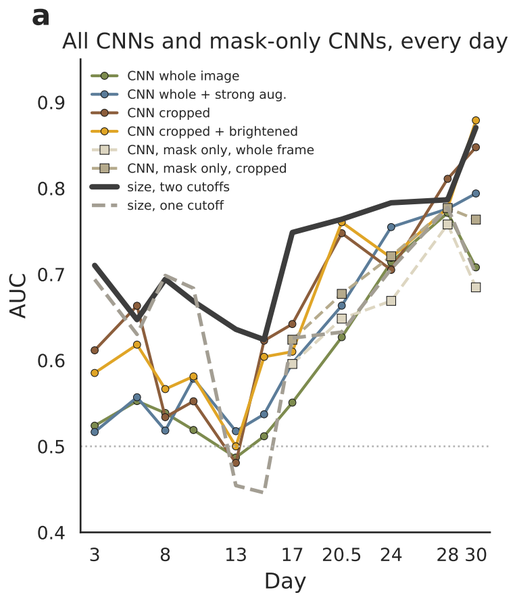
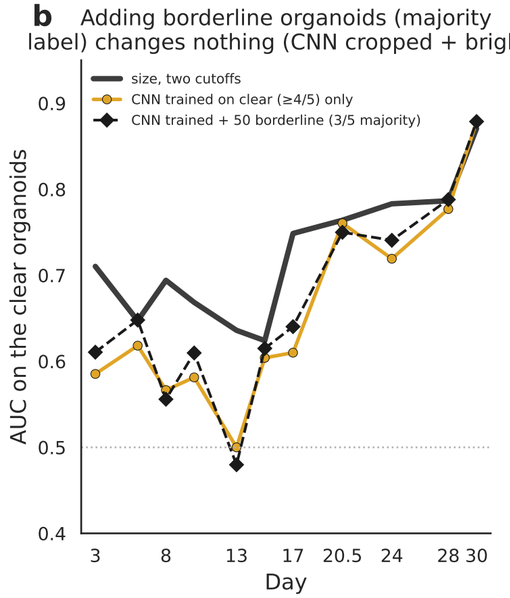
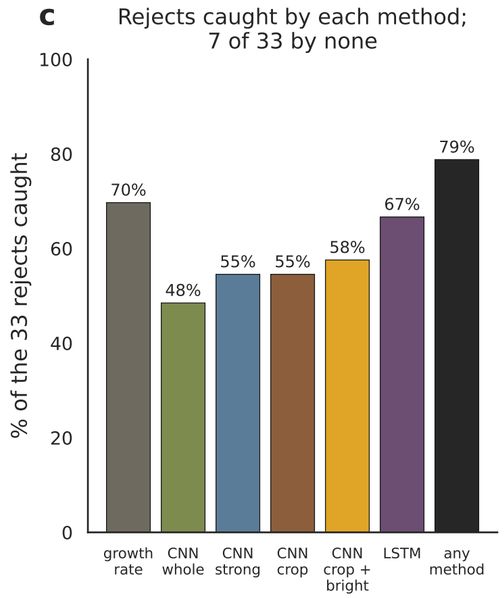
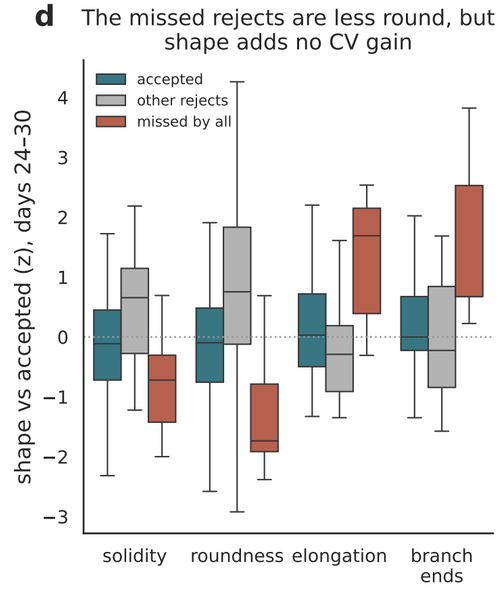
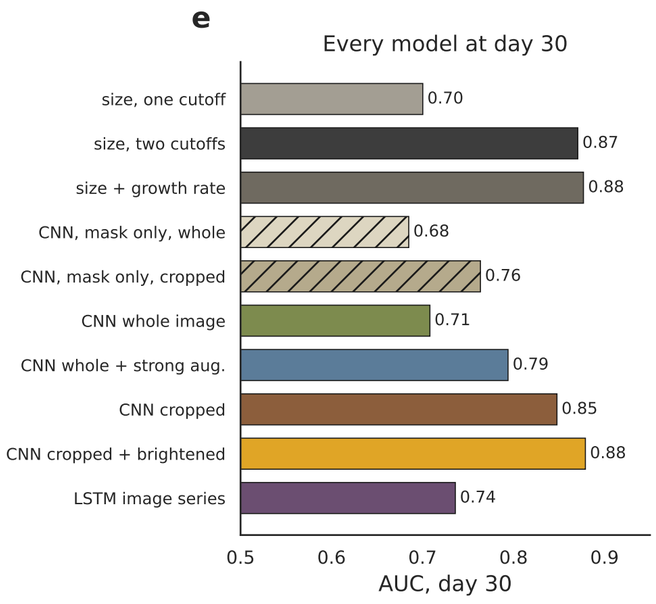
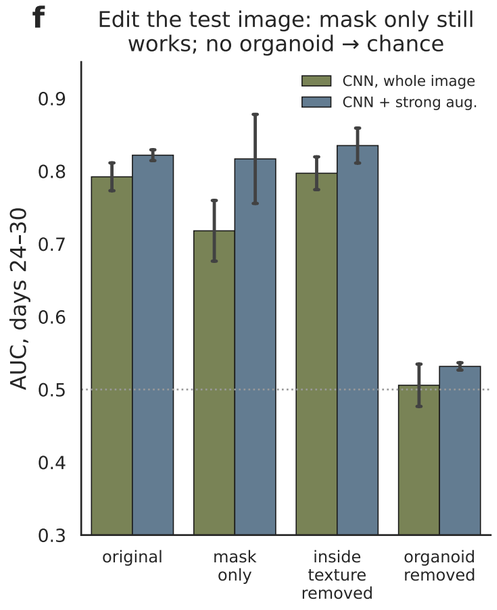

a
All CNNs and mask-only CNNs, every day
CNN whole image
CNN whole + strong aug.
CNN cropped
CNN cropped + brightened
CNN, mask only, whole frame
CNN, mask only, cropped
size, two cutoffs
size, one cutoff
AUC
Day

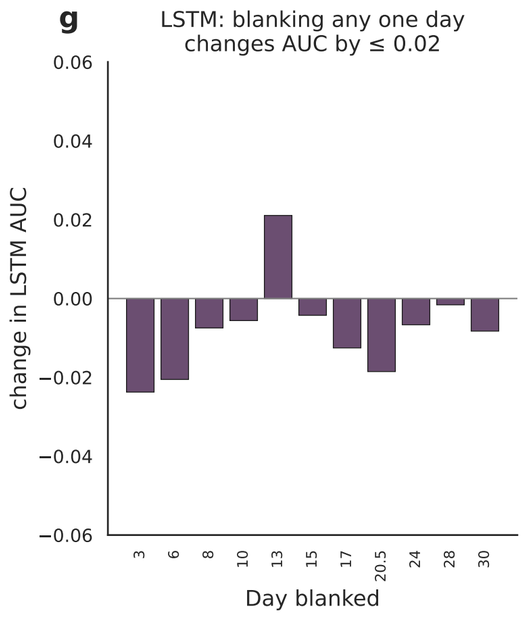

b
Adding borderline organoids (majority label) changes nothing (CNN cropped + bright.)
size, two cutoffs
CNN trained on clear (≥4/5) only
CNN trained + 50 borderline (3/5 majority)
AUC on the clear organoids
Day

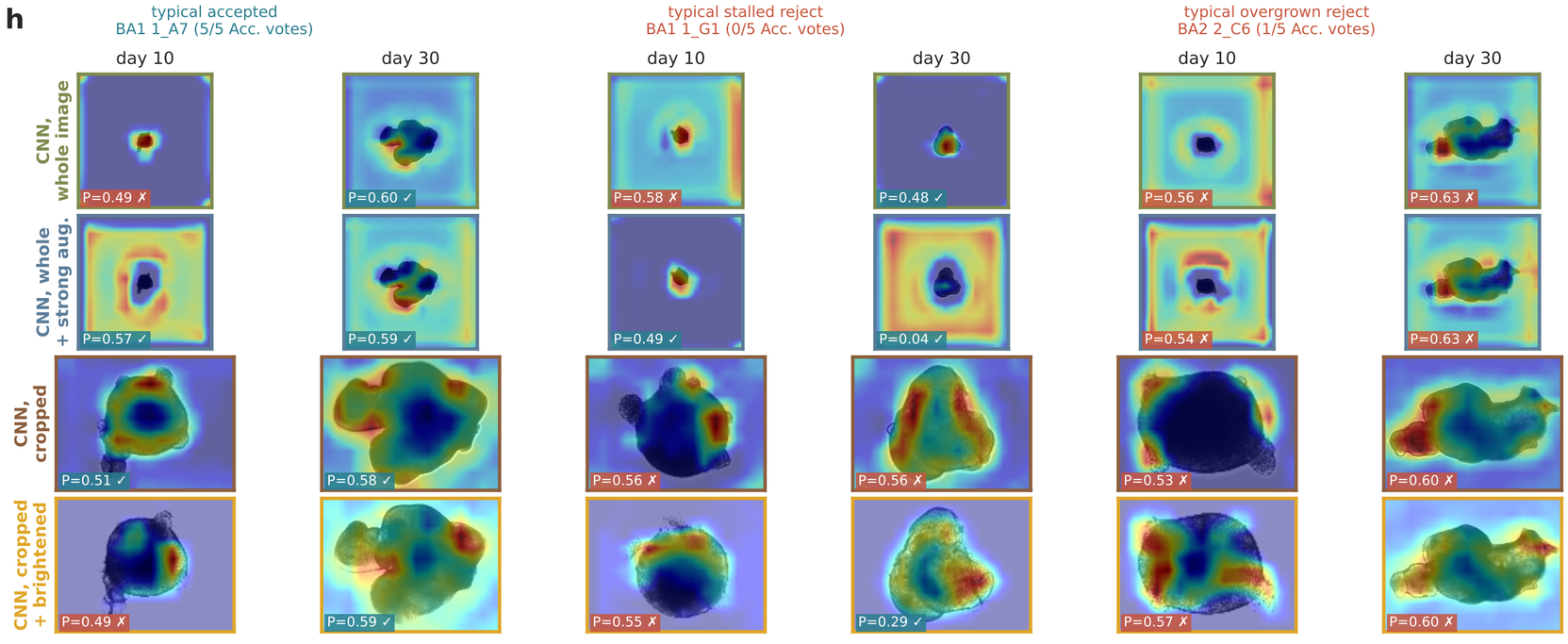

c
Rejects caught by each method; 7 of 33 by none
% of the 33 rejects caught
70%
48%
55%
55%
58%
67%
79%
growth rate
CNN whole
CNN strong
CNN crop
CNN crop + bright
LSTM
any method


d
The missed rejects are less round, but shape adds no CV gain
accepted
other rejects
missed by all
shape vs accepted (z), days 24–30
solidity
roundness
elongation
branch ends


e
Every model at day 30
size, one cutoff 0.70
size, two cutoffs 0.87
size + growth rate 0.88
CNN, mask only, whole 0.68
CNN, mask only, cropped 0.76
CNN whole image 0.71
CNN whole + strong aug. 0.79
CNN cropped 0.85
CNN cropped + brightened 0.88
LSTM image series 0.74
AUC, day 30


f
Edit the test image: mask only still works; no organoid → chance
CNN, whole image
CNN + strong aug.
AUC, days 24–30
original
mask only
inside texture removed
organoid removed


g
LSTM: blanking any one day changes AUC by ≤ 0.02
change in LSTM AUC
Day blanked


h
typical accepted
BA1 1_A7 (5/5 Acc. votes)
typical stalled reject
BA1 1_G1 (0/5 Acc. votes)
typical overgrown reject
BA2 2_C6 (1/5 Acc. votes)
day 10
day 30
CNN, whole image
CNN, whole + strong aug.
CNN, cropped
CNN, cropped + brightened
P=0.49 ✗
P=0.60 ✓
P=0.58 ✗
P=0.48 ✓
P=0.56 ✗
P=0.63 ✗
P=0.57 ✓
P=0.59 ✓
P=0.49 ✓
P=0.04 ✓
P=0.54 ✗
P=0.63 ✗
P=0.51 ✓
P=0.58 ✓
P=0.56 ✗
P=0.56 ✗
P=0.53 ✗
P=0.60 ✗
P=0.49 ✗
P=0.59 ✓
P=0.55 ✗
P=0.29 ✓
P=0.57 ✗
P=0.60 ✗